CLINICAL · EAR

Sudden sensorineural hearing loss

A patient who wakes with one ear suddenly dull, and a drum that looks completely normal, has an otological emergency until the tuning forks and an audiogram say otherwise. Treat early with steroids, then look for a cause: most cases are idiopathic, but a few hide a tumour, a stroke or an infection.

AT A GLANCE

- 30 dB over three frequencies within 3 days.
- Forks first; don't assume it's wax.
- Steroids early, intratympanic if needed.
- MRI for a retrocochlear cause.
- Half recover by 2 weeks, mostly with mild loss and no vertigo.

1 CHAPTER ONE Overview

Deafness overnight.

1.1 Definition

A sensorineural hearing loss of 30 dB or more over at least three contiguous frequencies, developing within three days.

1.2 Epidemiology

About 5 to 20 per 100,000 people a year, most often in middle age (median 40 to 54), equally in men and women. It is almost always one-sided.

1.3 Why it matters

Time matters: treatment works best in the first two weeks. And a minority have a serious underlying cause, including vestibular schwannoma and posterior circulation stroke.

2 CHAPTER TWO Causes

Usually idiopathic, but exclude the rest.

2.1 Idiopathic

Most cases have no identifiable cause. Three theories compete, and each may be right for some patients:

- **Viral:** cochleitis or neuritis, supported by seasonal clusters and viral serology.
- **Vascular:** thrombosis, embolism, spasm or haemorrhage in the labyrinthine artery, an end artery.
- **Membrane rupture:** of the cochlear membranes or the round window, letting perilymph leak.

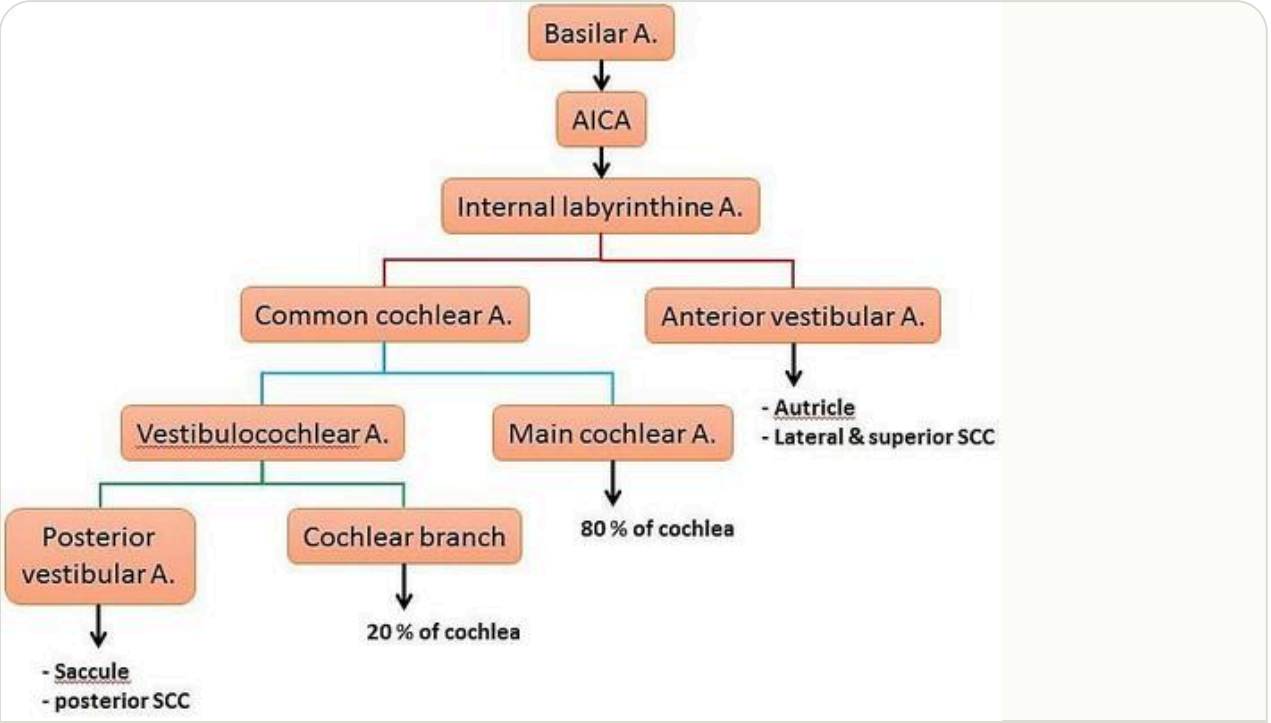


FIGURE 1 Branches of the labyrinthine artery, an end artery, which explains the vascular theory.
Anatomy atlas · ENT Steps p. 82

2.2 Identifiable causes

CAUSES TO EXCLUDE

- INFECTION **Mumps, measles, herpes zoster** meningitis, syphilis, acute otitis media, HIV
- TRAUMA **Head injury, ear surgery** blast, barotrauma, perilymph fistula
- VASCULAR **Stroke (AICA)** leukaemia, sickle cell, polycythaemia, embolism, bypass surgery
- OTOLOGICAL **Ménière's disease** Cogan syndrome, enlarged vestibular aqueduct
- TUMOUR **Vestibular schwannoma** other cerebellopontine angle tumours, metastases
- OTHER **Ototoxic drugs** multiple sclerosis, hypothyroidism, sarcoidosis, autoimmune disease, non-organic loss

PERILYMPH FISTULA

A leak at the oval or round window may follow stapes surgery, barotrauma, heavy lifting or straining. Spontaneous fistula is possible but probably over-diagnosed.

3 CHAPTER THREE Clinical features

One ear, often on waking.

3.1 Symptoms

3.1.1 Hearing loss

Sudden, one-sided, often noticed on waking or over a few hours. Patients frequently describe it as a blocked ear.

3.1.2 Tinnitus and fullness

Often precede or accompany the loss.

3.1.3 Vertigo

> Present in some, and a sign of a worse prognosis.

3.2 Signs

The canal and drum are normal. Tuning forks are the key bedside test: Weber lateralises to the *good* ear, and Rinne remains positive in the affected ear.

NOT WAX UNTIL PROVEN OTHERWISE

A blocked ear after a cold, a swim or a flight can be sudden sensorineural loss. If you assume wax or fluid without doing forks, the window for treatment can close.

4 CHAPTER FOUR Investigations

Confirm, then search.

4.1 Audiometry

Pure-tone audiogram with masking, and speech discrimination. The pattern (low-frequency, high-frequency, flat or total) helps prognosis.

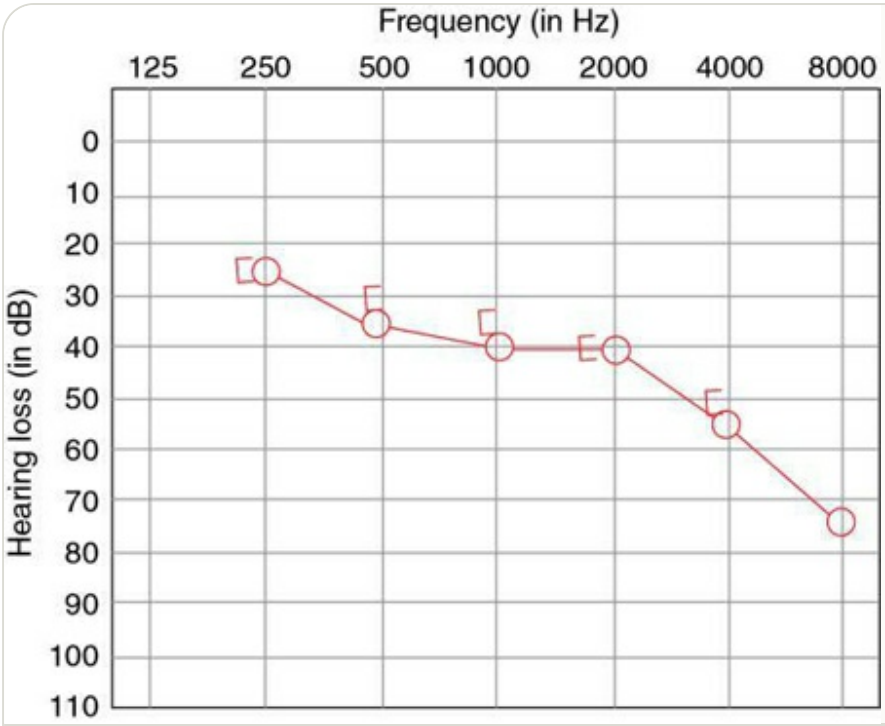


FIGURE 2 Audiogram of a right ear with sensorineural loss: air and bone conduction fall together, with no air-bone gap.

Dhingra, fig. 5.6

4.2 Tympanometry

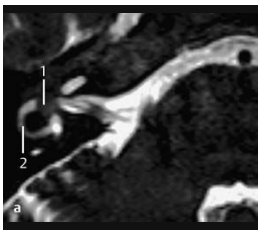
Normal, which excludes middle ear fluid.

4.3 MRI

MRI of the internal auditory canals and cerebellopontine angles with gadolinium, to exclude vestibular schwannoma and other retrocochlear lesions. If MRI is impossible, auditory brainstem response.

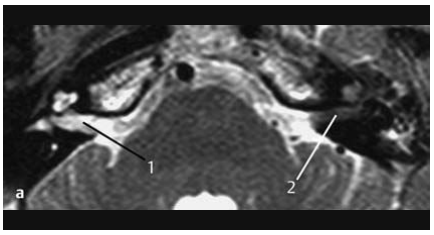
IMAGING IN THE APP 3 cases

Retrocochlear and labyrinthine causes: schwannoma, labyrinthitis and otosyphilis.



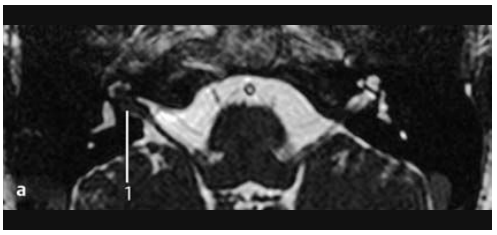
Vestibular Schwannoma

Tumours · 13 images



Labyrinthitis and Subclinical Meningitis

Infection & inflammation · 5 images



Otosyphilis and Intracranial Complications

Infection & inflammation · 6 images

4.4 Blood tests by history

Not routine for everyone. Full blood count, ESR, glucose, lipids, thyroid function and syphilis serology when the history suggests them; autoimmune screens for bilateral or fluctuating loss.

4.5 Exploration

Exploratory tympanotomy only when a fistula is strongly suspected, after trauma or straining.

4.6 Differential diagnosis

- Wax or middle ear effusion: forks and tympanometry.
- Ménière's disease: fluctuation, low-frequency loss, attacks of vertigo.
- Vestibular schwannoma: MRI.
- Anterior inferior cerebellar artery stroke: vertigo, ataxia, facial weakness.
- Non-organic hearing loss: inconsistent results, objective tests normal.

5 CHAPTER FIVE Management

Early steroids, then rehabilitation.

5.1 On the day

SUDDEN HEARING LOSS IN CLINIC OR THE EMERGENCY DEPARTMENT

1. Do the forks: Weber to the good ear, Rinne positive in the bad ear.
2. Look for wax; don't accept a blocked ear without testing.
3. Arrange an audiogram the same or the next day.
4. Offer steroids early.
5. Facial weakness, vertigo with ataxia, or vascular risk factors: treat as a posterior circulation stroke.

5.2 Steroids

5.2.1 Oral

High-dose prednisolone, for example 1 mg/kg (up to about 60 mg) a day for a week, then tapered. Dhingra gives prednisolone 40 to 60 mg each morning for a week, tapered over three weeks.

5.2.2 Intratympanic

Dexamethasone or methylprednisolone injected through the drum. Used first-line when systemic steroids are unsafe (poorly controlled diabetes, for example), or as salvage when oral treatment has failed.

5.3 Other treatments

Antivirals have no proven benefit. Older adjuncts, such as bed rest, carbogen inhalation, vasodilators, dextran, a low-salt diet with diuretics and hyperbaric oxygen, have not clearly beaten natural recovery. Dextran is contraindicated in heart failure and bleeding disorders.

WHERE THE BOOKS DIFFER · ADJUNCT TREATMENTS

Dhingra
Lists carbogen, low salt and hyperbaric oxygen among the options, while noting that no treatment has proven better than spontaneous recovery.

Lee
Recommends steroids, orally or intratympanically; finds no evidence for antivirals.

5.4 Rehabilitation

If recovery is poor: a hearing aid for a partial loss, or a CROS aid, bone-conduction device or cochlear implant for single-sided deafness. Tinnitus counselling as needed.

OPERATIVE GUIDES IN THEATRE 2 operations

Cochlear Implantation in Adults
Implant for lasting profound loss

Acoustic Neuroma
If a schwannoma is found

6 CHAPTER SIX Prognosis

What predicts recovery.

About half recover by two weeks; recovery after a month is unusual.

PROGNOSIS IN SUDDEN HEARING LOSS

FACTOR	BETTER	WORSE
Severity	Mild	Severe or total
Frequencies	Low or mid	High or flat
Vertigo	Absent	Present
Age	Younger	Older
Treatment	Early	Late
Early recovery	Within two weeks	None by two weeks

6.1 Follow-up

Repeat the audiogram during treatment and at the end. Counsel the patient to protect the other ear from noise and ototoxic drugs.

7 CHAPTER SEVEN For the examination

What the examiner wants to hear.

7.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **The definition:** 30 dB, three frequencies, three days.
- **Tuning forks:** Weber to the good ear, Rinne positive.
- **Theories:** viral, vascular, membrane rupture.
- **MRI** to exclude a schwannoma.
- **Steroids,** oral and intratympanic.
- **Prognostic factors.**

7.2 Common pitfalls

COMMON PITFALLS

- Calling it wax or a cold without doing forks.
- Delaying treatment while waiting for an audiogram slot.
- Forgetting MRI once the acute phase has passed.
- Missing a stroke in a patient with vertigo and ataxia.

Written for this app from Oxford Textbook of Otolaryngology ch. 11, Oxford Specialist Handbook ch. 18, Oxford Handbook of ENT ch. 7, Dhingra ch. 5 and K. J. Lee ch. 23. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

On imaging

The imaging of this topic: key findings, differentials and representative scans.

INNER EAR & IAC · TUMOURS

Vestibular Schwannoma

IMAGING FINDINGS

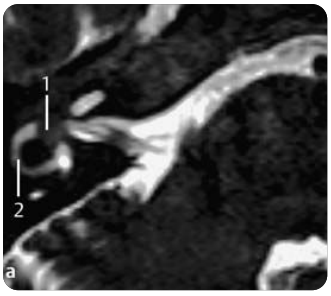
- Enhancing mass centred on the porus, "ice cream on a cone"
- Dark filling defect in bright CSF on high-resolution T2; fundal cap of fluid
- Acute angle with the petrous bone; widened porus; cystic change in larger tumours

TOP DIFFERENTIALS

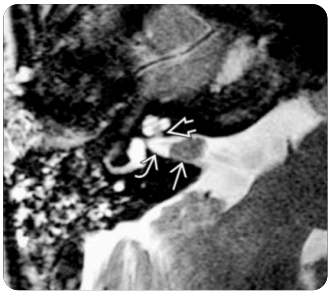
- Meningioma (broad dural base, dural tail)
- Epidermoid cyst (DWI bright)
- Facial nerve schwannoma (labyrinthine segment)

CHECKLIST

- Size in canal and cistern; brainstem and fourth ventricle (Koos)
- Fundal cap and cochlear nerve canal for hearing preservation



MRI, T2-weighted, axial.A partial lack of fluid in the vestibule and horizontal semicircular canals is evident (1). A mass lesion is present in the vestibul…



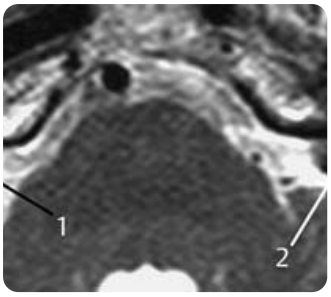
Axial T2WI MR reveals a small intracanalicular vestibular schwannoma visualized as a soft tissue intensity mass surrounded by…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1222.

INNER EAR & IAC · INFECTION & INFLAMMATION

Labyrinthitis and Subclinical Meningitis

- Otoscopy may exclude middle ear infections although CT may be indicated in some cases. MRI will show (retro)cochlear enhancement in cases of (acute) infections.
- Inflammatory involvement of the intralabyrinthine structures may result in fibrosis and ossification. In bilateral cases, this may hamper insertion of a cochlear implant.

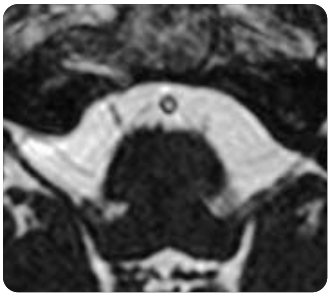


MRI, T2-weighted, axial. Note the normal fluid content of the internal auditory canal on the right side (1), whereas on the affected side, the…

INNER EAR & IAC · INFECTION & INFLAMMATION

Otosyphilis and Intracranial Complications

- Cranial nerve deficits, rapid progression, and multiple lesions are strongly indicative of inflammatory disorders. If malignancy is suspected, beware of involvement of other organs as primary sites of possible metastases.
- Inflammatory involvement of the intralabyrinthine structures may result in fibrosis and/or ossification. In bilateral cases, this may hamper insertion of a cochlear implant, although its success may also be restricted by retrocochlear deficits.



Two weeks prior to this MRI investigation, this young man presented with sudden deafness of the right ear, followed by severe…